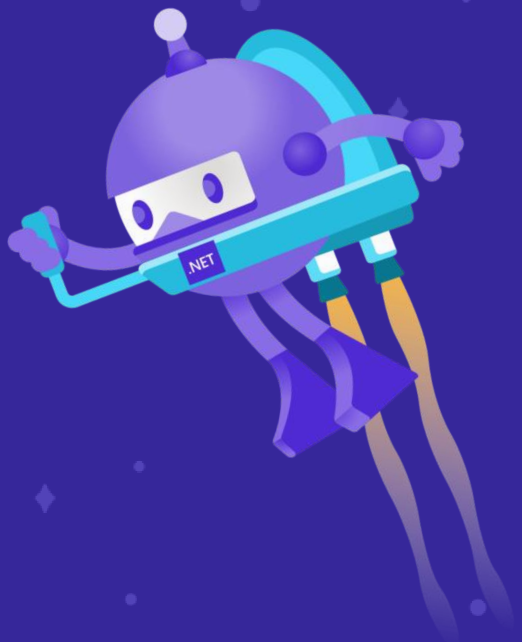
# What’s new in F# 5 & 6

Don Syme |> @dsymetweets



# Agenda

- One pandemic, two language versions
- Changes in how we talk about F#
- Walk through F# 5 and F# 6

# An update on how we talk about F#

# A marketing exercise we did

- Don Syme, Phillip Carter, Niklas Gustafsson also Reed Copsey, Isaac Abraham and others
- Target markets:
  - **Python programmers** <u>hitting the limits</u> w.r.t typing and performance
  - **C# programmers** wanting succinct, data-oriented programming <u>without leaving the .NET ecosystem</u>
  - Plus Scala, Julia, Go, Rust and everything else ☺

# A marketing exercise

## Methodology

- **Write down all the technical features/qualities of F#**
- **Mark them for relevance to the target market**
- **Choose the top three**

# A marketing exercise, for example:

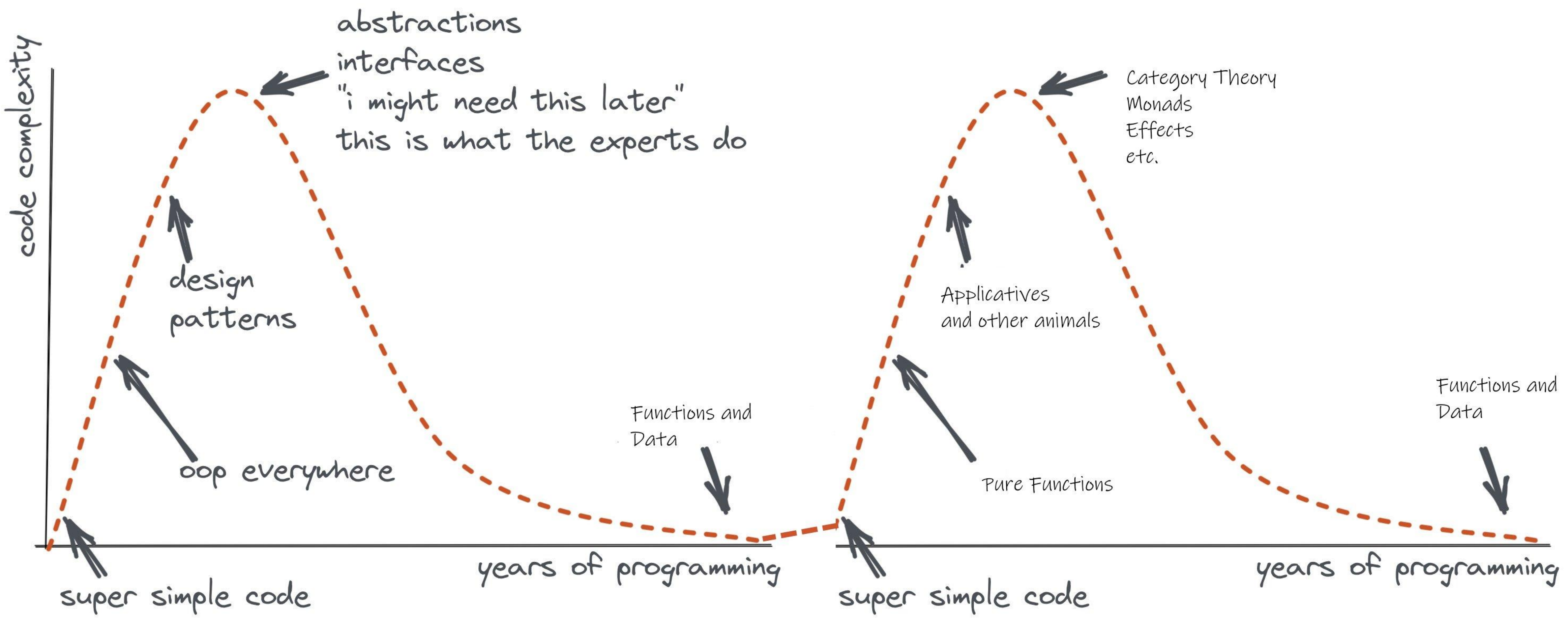
| Quality | Features | Python devs hitting limits | C# devs wanting low ceremony |
|---|---|---|---|
| **Succinctness** | Type inference<br>Expression-oriented<br>Tuples<br>Active patterns<br>…. | Table stakes ✔ | Appeals! ✔ ✔ ✔ |
| **Robustness** | Types<br>No-nulls<br>Functional abstraction<br>… | Appeals! ✔ ✔ ✔ | Table stakes ✔ |
| **Performance** | Typed code<br>Generics without boxing<br>…. | Appeals! ✔ ✔ ✔ | Table stakes ✔ |
| **Expressiveness** | … | … | … |
| **Interoperable** | … | … | … |

# Chosen Themes

- Succinct
- Robust
- Performant

# Remember the eternal sweet spot of programming!

## Functions + Data + Types



# What is F# good at?

dot.net/fsharp

F# is good for **programming**

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F# is good for **web programming**

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F# is good for **cloud programming**

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F# is good for **AI/ML/data-science programming**

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F# is good for **succinct programming**

dot.net/fsharp

F# is good for **correct programming**

dot.net/fsharp

F# is good for **performant programming**

dot.net/fsharp

F# is good for **programming**

dot.net/fsharp

# F# get started

```
dotnet new -lang F#

dotnet build
```

F# tools are part of the .NET SDK, available everywhere

.NET

www.dot.net

# F# for the web backend

```
dotnet new -i "giraffe-template::*"

dotnet giraffe
```

High perf, functional server-side programming

GIRAFFE

A functional ASP.NET Core micro web framework for building rich web applications.

github.com/giraffe-fsharp/Giraffe

# F# for the web frontend (JavaScript)

```
dotnet new -i "Fable.Template::*"

dotnet new fable
npm install
npm start
```

You can use F# as a JavaScript language

websharper.com

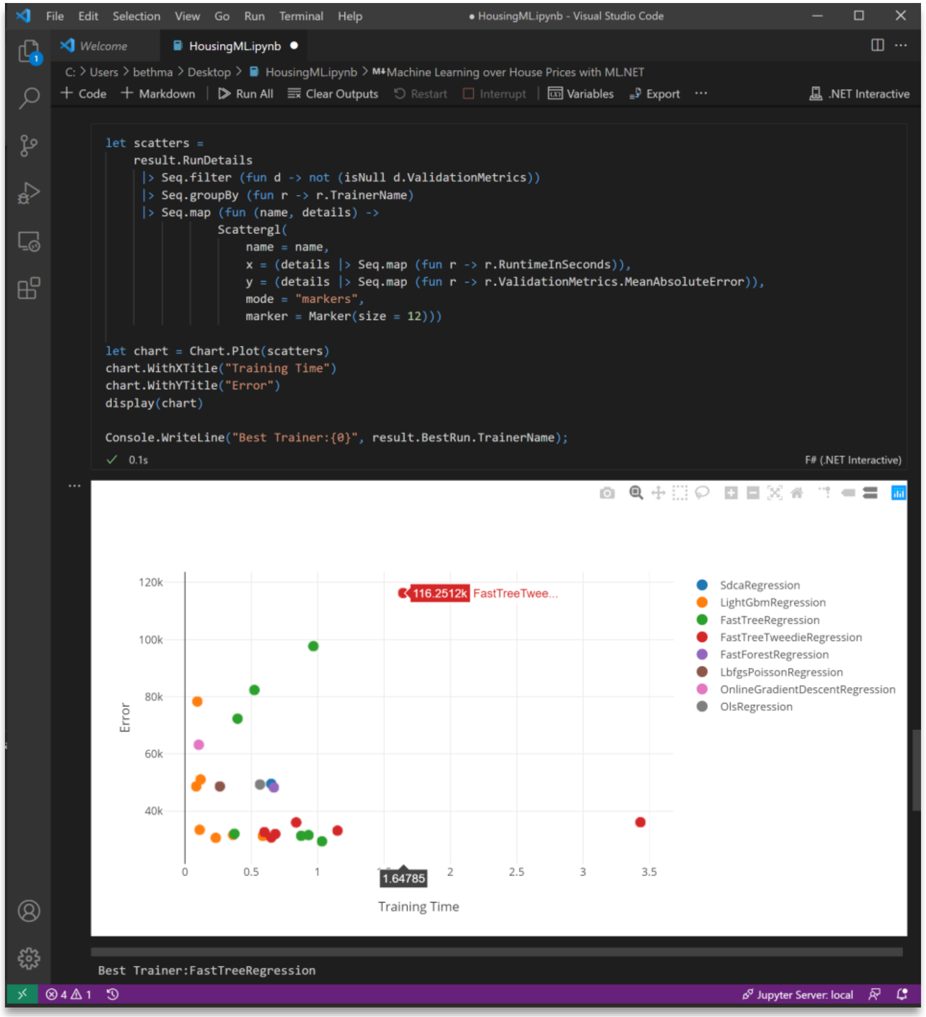
fable.io

# F# for the full stack with SAFE

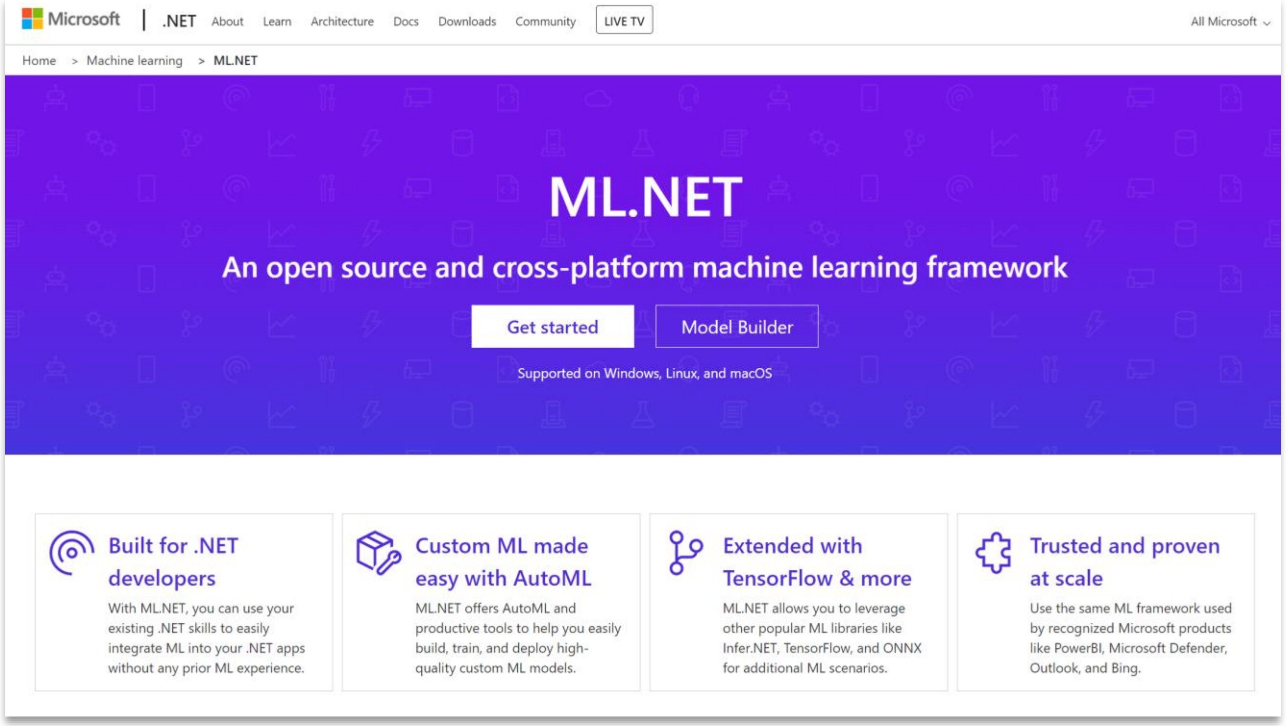
```
dotnet new -i SAFE.Template

dotnet new SAFE
dotnet tool restore
dotnet run
```

safe = GIRAFFE + FABLE + Microsoft Azure

# F# for data science & ML




```
#r "nuget:Microsoft.ML,1.4.0"
#r "nuget:Microsoft.ML.AutoML,0.16.0"
```




github.com/dotnet/interactive

dot.net/ml

# F# 5! Release Nov 2020

# F# 5 - Laundry list

- FSharp.Core targets netstandard2.0
- **#r "nuget: …“**
- .NET Interactive
- **String interpolation!**
- nameof
- **“open type”**
- Slicing updates
- **let! .. and! ..**
- Stack traces for async
- Multi generic instantiations
- Default interface member consumption
- Interop with nullable value types
- **Improved Map/Set performance**
- Improved Compiler performance
- **Checked XML docs**

# F# 6!   Release Nov 2021

# F# 6 - Laundry list

- **task { … }**
- Resumable code
- **expr[idx]**
- Structs for partial active pattern returns
- Overloaded custom operations in CEs
- “as” patterns
- Indentation syntax revisions
- InlineIfLambda
- **4x perf for list and array expressions**
- Additional implicit conversions
- Immutable collection updates
- More unit annotations
- **! and := soft deprecation**
- Remove legacy features
- **Pipeline debugging**
- Value shadowing
- **Tooling perf, scalability (64-bit, multi-thread)**
- **.NET Core default for VS scripting**
- **Scripting respects global.json**
- General improvements in .NET 6
- General improvements in Visual Studio 2022

# String interpolation

- The very first F# RFC FS-1001, from 2014
  - Improved readability
  - Reduced errors
  - More succinct
  - Less to learn
- Design has been iterated, very happy with the result
- Safer than any other language with string interpolation
- Aim
  - Align with C#
  - But keep strong typing by unifying with printfn
  - Allow use with printfn + friends
- Demo

# “open type”

- RFC FS-1068
- Increase expressivity of F# DSLs and ultra-succinct mode programming
  - Single-word “functions” can be overloaded members accepting named, optional, ParamArray arguments
- Demo
- **Caveat emptor**. Don’t inflict this on yourself or your friends without real thought about ramifications

# #r "nuget: ..."

- People rate this as the #1 feature for Python programmers
- Combined with **global.json** in script directory allows you to "lock down" F# scripts as trustworthy

- #r "nuget: Newtonsoft.Json"
- #r "nuget: Newtonsoft.Json, 11.0.1"

- #i "nuget: https://my-remote-package-source/index.json"

# let! … and! ….

- Extends F# computation expressions

# Computational modalities in F# (computation expressions)

Iterator methods (from Icon)

Query expressions

Comprehensions

Async

Async sequences

**modality { code }**

2007 – monads, monoids, async, comprehensions
2012 – queries, extended comprehensions

# Computational modalities in C#, F# (computation expressions)

C# Iterator methods

C# Query expressions

C# Collection initialization

C# Async methods

C# Async iterator methods

**seq { code + yield }**

**query { code + yield + custom }**

**[ code + yield ]**

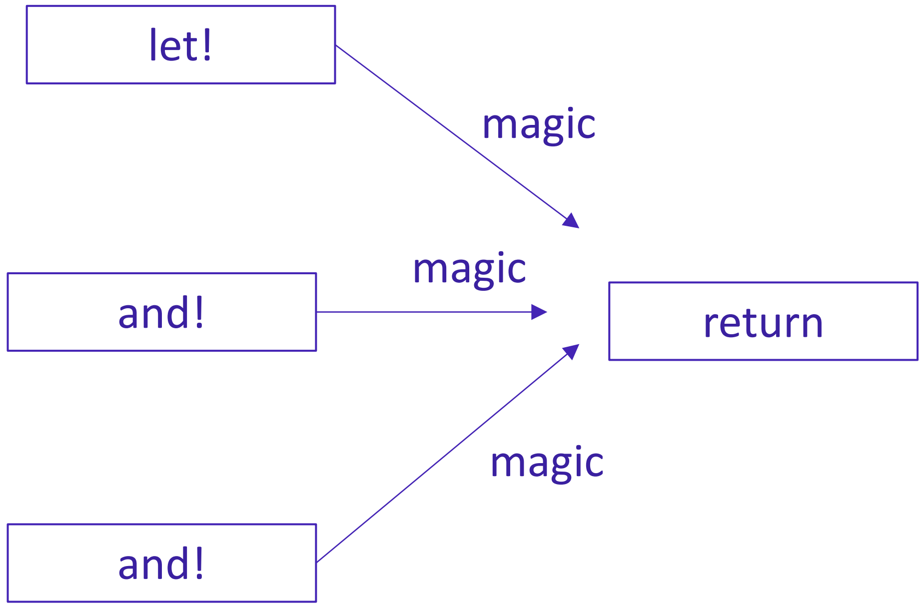
**async { code + let! }**

**asyncSeq { code + let! + yield }**

**coroutine { code + yield() }**

# Monadic binding (let!)

# Applicative binding (and!)



# let! … and! ….

- RFC FS-1063
- Directly from the last F# London meetup before pandemic ☺
- “**let! … and! …**” in computations
  - Maps to overloaded BindReturn/MergeSources
  - See RFC
- Used for **applicatives** - two-phased computations
  - Separate “compose” from “run”
  - Static “composition”, then dynamic “run”
  - Example: graphs of fixed size (pre-allocate buffers, then run)
  - Example: validation  (compose validators, then run)
- Demo

# expr[idx]

- RFC FS-1110
- Why?
  - We lose people for no reason
  - “Making F# better by moving it towards it already was”
- Demo
- Caveats

# task { .. }

- Why, what
- Demo
- Differences with async
  - A task is a single mutable, awaitable register
  - Hot start
  - No implicit cancellation checks
  - No implicit cancellation propagation
  - No tailcalls
- Use primarily for interop when heavily consuming or producing tasks
- Debugging can be better
- Builds on “resumable code”, has other applications

# !, :=, incr, decr soft deprecation

- RFC FS-1111
- These operators are highly confusing to beginners
- They are now rarely used
- Soft deprecation via informational messages
- Demo

# Performance and Scalability

- F# Compiler Service new multi-threaded (no reactor queue!)
  - Makes IDEs MUCH more responsive
- Visual Studio 2022 now 64-bit
- Other major performance improvements, see announcements

# Pipeline Debugging!

- Demo
- Also shadowing on values shown in debugging

# Performance for [ .. ] and [| ... | ]

- Demo/link

# Improved Map/Set performance

- By Victor Baybekov
- In FSharp.Core 5.0.0+

For `Map`, the improved operations are:

- Retrieval via index – up to 50% faster
- `containsKey` – up to 50% faster
- `count` – up to 50% faster
- `iter` – up to 15% faster
- `add` – up to 40% faster
- `remove` – up to 33% faster

For `Set`, the improved operations are:

- `containsKey` – up to 40% faster
- `isSubsetOf` – up to 40% faster
- `max` – up to 50% faster
- `count` – up to 50% faster
- `add` – up to 30% faster
- `remove` – up to 15% faster

# Doc improvements

- XML Doc checking!
- FSharp.Core examples! (500)
- docs.microsoft.com/dotnet/fsharp/ revamp!

- Call to action. Stackoverflow is becoming a problem for us. Please update your answers!

# Huge thank you to contributors

- https://devblogs.microsoft.com/dotnet/whats-new-in-fsharp-6/#thanks-and-acknowledgments

# Thanks!